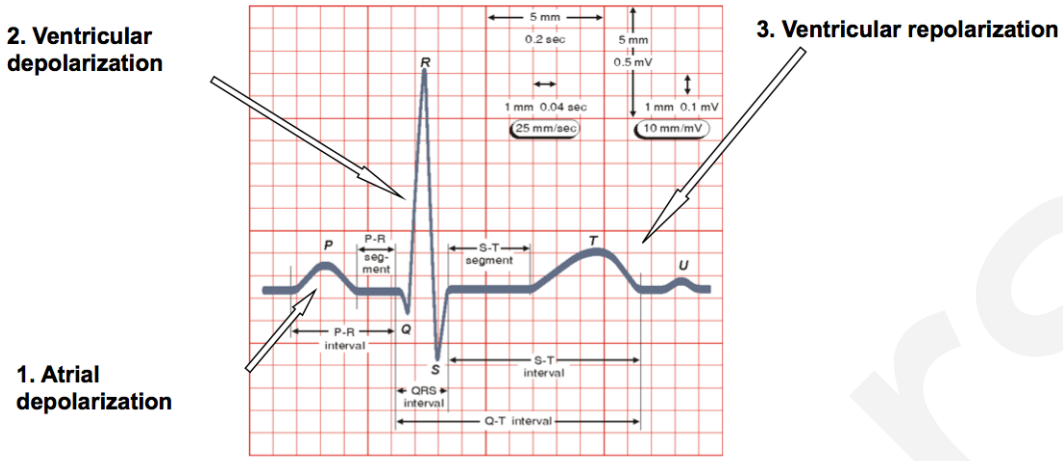
1. This question relates to the design of an instrument for measuring bio-potentials.

a) Sketch a typical ECG waveform labelling each segment. Explain briefly what each segment of the ECG waveform represents during a single heartbeat.




P wave represents the normal atrium (upper heart chambers) depolarization;
QRS complex (one single heart beat) corresponds to the depolarization of the right and left ventricles (lower heart chambers);
T wave represents the repolarization (or recovery) of the ventricles.
[2 marks for drawing with labels and 2 marks for brief explanations]

[4]

b) Table 1.1 lists five commonly observed bio-potentials and their amplitude and frequency characteristics. A doctor requires an instrumentation system to record the ECG from the heart. The front end will use the instrumentation amplifier shown in Figure 1

Design a complete instrumentation system for recoding ECG from the chest according to the following specifications:

Set Instrumentation Amplifier Gain: 200
State all assumptions and values of resistors and capacitors used. You may assume that the instrumentation system utilizes the whole dynamic range available and 10nF capacitors are available for the filters.

[8]

$V_{OUT} = -(V_1 - V_2)(1 + 2R_2/R_1)(R_4/R_3)$
For gain of 200 we select R4=R3= 20 KΩ and R2= 199 KΩ, R1= 2 KΩ
[1 mark]

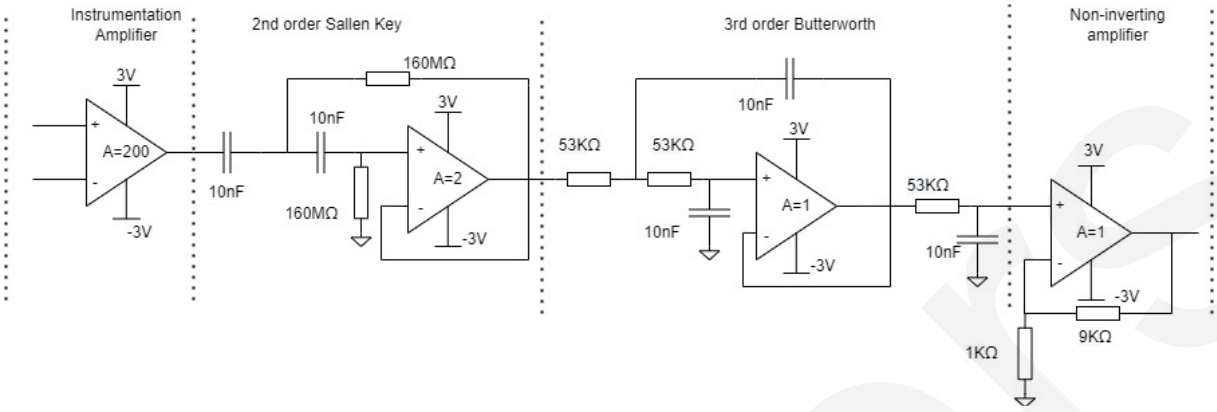
Low-Pass Filter Roll off: 60dB/decade
2nd order sallen key filter + LPF
Low pass value $R_3=1/2\pi C_2 f_{LPF} = 1/2\pi(10n)(300Hz)=53K\Omega$ [1 mark]
High-Pass Filter Roll off: 40dB/decade
Two First order RCs
High pass value $R_2=1/2\pi C_1 f_{HPF} = 1/2\pi(10nF)(0.1Hz)=160M\Omega$ [1 mark]

Supply Range: -3V to 3V
VDD=3V, GND=0V, VSS=3V
Max ECG = 3mV (from table)
$V_{inptp}$ x A1 x A2=$6V_{ptp}$

Answers

A1=200
3mVx200xA2=6V
A2= 10(gain of filter and final stage) [1 mark]
Realised with a non-inverting amplifier with R1=9KΩ and R2=1KΩ [1 mark]

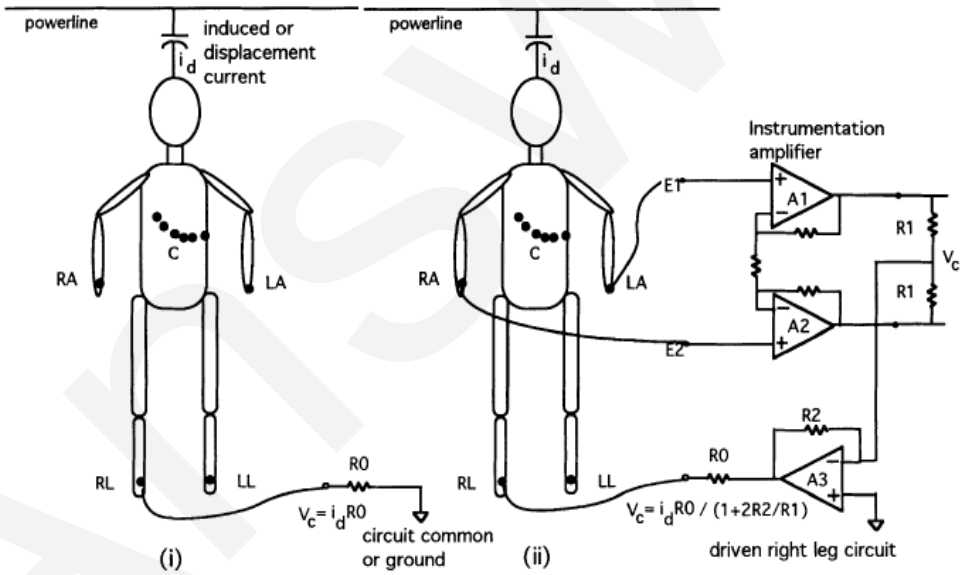
Final Circuit:




Complete circuit with annotated component values [4marks]

c) Design a circuit to reduce the common mode voltage created by current capacitively coupled to the body from the power-lines and derive the equation showing how this improvement has an effect.

Modification of instrumentation amplifier:




[2 marks for circuit]

ii) Show that the common mode voltage with this circuit reduces to:

$$V_{CM} = \frac{i_D R_0}{1 + 2\frac{R_2}{R_1}}$$

Derivation from notes:

$$\frac{V_{cm}}{R_1/2}+\frac{V_{out}}{R_2}=0$$

$$V_{out}=-\frac{2R_2}{R_1}V_{cm}$$

$$V_{cm}=R_0 i_d+V_{out} \Rightarrow \frac{R_0 i_d}{1+2R_2/R_1}$$

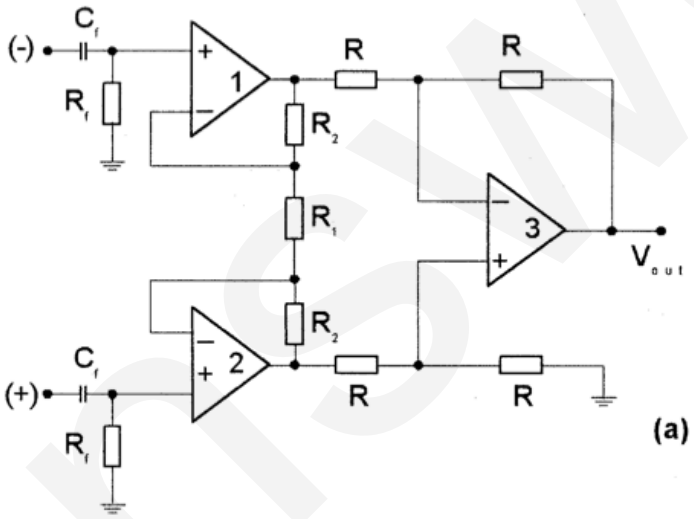
[2 marks]

d) After measurement of the ECG you notice that the instrumentation amplifier keeps saturating due to a large DC offset voltage between the input electrodes. The impedance of the input electrodes is 1MΩ. Sketch a modified instrumentation amplifier able to solve this problem,

Sketch of AC coupled Instrumentation Amplifier. Need to make sure HPF resistor values are 10x larger than impedance of electrodes. Previous calculation satisfies this criterion and thus the same RC values can be used. [Assumption 1 mark]

High pass value $R_f=1/2\pi C_f f_{HPF}$ = $1/2\pi(10nF)(0.1Hz)=160M\Omega$ [1 mark]

Circuit [2 marks]




[4]

2. This question relates to the design of a deep brain stimulator (DBS):

a) Discuss the two main requirements of a stimulation circuit. [4]

Firstly, a stimulation circuit must precisely be able to control the amount of charge delivered to the load (i.e. target tissue) [1]. This is typically achieved by employing current, voltage or charge controlled stimulus generator [1]. Secondly, a stimulation circuit must ensure the net charge remains near zero after a stimulation cycle is complete, i.e. cathodic and anodic charge is equal and opposite [1]. In case there is a net charge, some compensation method is needed, for example, a shorting phase [1].

b) Why do DBS electrodes typically have four electrode contacts? [3]

This is to ensure the target brain region can be precisely targeted [1] – providing a means for fine tuning after the electrode lead is implanted (e.g. this can compensate for any slight positioning error) [1]. It is

Answers

also possible to employ a tripolar or multipolar electrode configuration aiming to shape or "steer" the stimulation focus. [1]

A typical DBS system needs to generate a symmetric biphasic stimulus in range 0.5-2.5mA, with pulse width in range 60-250µs, at a repetition rate of 125-500Hz.

c) Estimate the device lifetime for a DBS implant with 6.3Ah primary cell battery. If you were to develop a rechargeable version of this device, what rechargeable battery capacity would you select? Explain your rationale. [7]

- Minimum average current consumption = 125Hz x 60µs x 0.5mA = 3.75µA, Maximum average current consumption = 500Hz x 250µs x 2.5mA = 312.5µA [1]
- These values assume no losses and 100% efficiency, i.e. current consumption is all stimulation current. [1]
- Assuming 25% device efficiency: average current consumption = 15µA to 1.25mA. [1]
- Therefore, battery lifetime = 6 months to 48 years (depending on settings). [1]
- If we were to pick a rechargeable battery, perhaps design for recharge once every 2 weeks. [1]
- Therefore battery capacity should be 15µA x 24 x 14 = 5mAh to 1.25mA x 24 x 14 = 420mAh (depending on settings). [1]
- There would be a strong desire to reduce battery due to device size/weight. [1]

d) Sketch a possible design (system architecture, i.e. block diagram) for this DBS device. [6]

Key blocks to include (in 3 categories):
1. Battery, power management circuits, telemetry circuits (for programming and/or recharge) [1.5]
2. Digital circuits: patient settings (memory), controller, crystal oscillator/clock. [1.5]
3. Stimulator circuit: including DAC, V-to-I (or iDAC), Current buffer, 2:4 demultiplexer, 4-electrode contacts, capacitively coupled outputs. [3]

Answers

3. This question relates to the design of a novel integrated sensor for chemical measurement.

a) Electrochemical measurement has as a basis a redox reaction. Explain what is meant by a redox reaction using reaction equations where necessary.

Bookwork

Definition of reduction with equation [2 marks]

Definition of oxidation with equation [2 marks]

b) Figure 1 shows a new pixel architecture to be used a part of an array in an integrated CMOS based diagnostic device. The total pixel size is 150um x150um with a total sensing area of 100um x 100um which is capable of sensing both glucose and pH. The working electrode is coated with glucose oxidase to give a peak current to glucose at $V_{Cell}$=0.7 V according to the following reaction:

$$\text{glucose} + O_2 \xrightarrow{\text{glusoce oxidase}} \text{gluconic acid} + H_2O_2 \quad (1)$$

$$H_2O_2 \rightarrow O_2 + 2H^+ + 2e^-$$

i) The sensor shall be used to measure a physiological glucose range of 0-20mM. The peak redox current for a scan rate of 1V/s is given by the following equation:

$$i_p = 2.69 \times 10^5 n^{3/2} A D^{\frac{1}{2}} C$$

where the diffusion coefficient D=4.38X10$^{-8}$ m$^2$/s. Calculate the maximum range of the WE current, using the dimensions shown in Figure 1, to capture the physiological range of glucose.

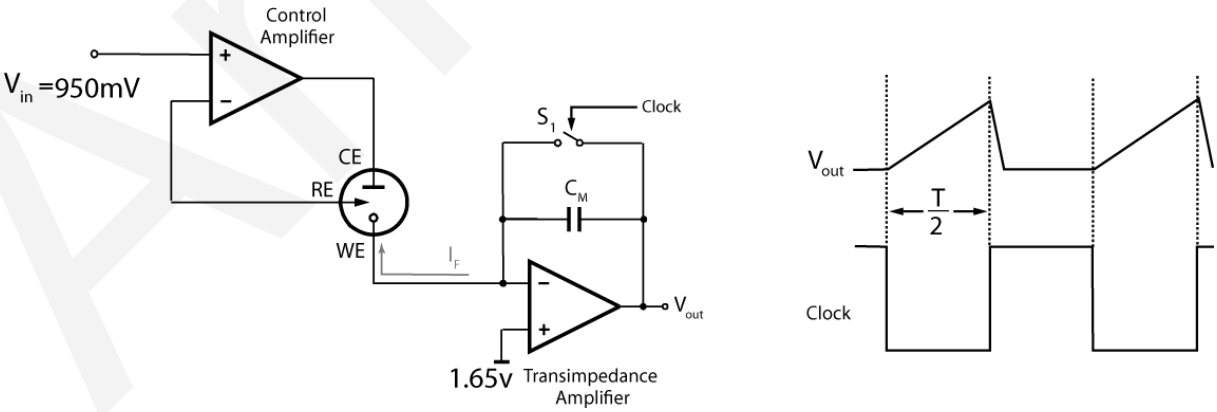
Total area of WE=50umx20um=1000um$^2$ [1 mark]

Number of electrons =2

Peak current at 20mM

Ip=2.69x10$^5$x2$^{3/2}$x50x10$^{-6}$ x 20x10$^{-6}$ x(4.38x10$^{-8}$)$^{0.5}$x20x10$^{-3}$x1000=3.18uA [1 mark]

ii) The amperometric sensor in the pixel shall be driven by a potentiostat and transimpedance amplifier which uses an integration capacitor, CE to convert the sensed current to a voltage. The circuit operates of a single supply with $V_{DD}$=3.3V and $V_{SS}$=0V. Sketch the schematic of a suitable measurement system used to interface to the amperometric sensor and label the voltages of all electrodes




Potentiostat [2marks]

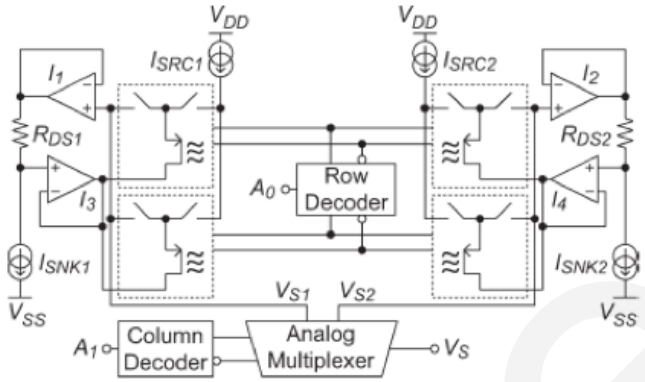
Tansimpedance amplifier [2 marks]

Correct voltages [2 marks]

iii) The transimpedance amplifier has a poly-silicon integration capacitor determined by the remainder of the area in the pixel shown in Figure 1. A capacitance per unit area of poly-silicon is 4x10$^{-4}$ pF/um$^2$. Calculate a suitable clock frequency that maximizes the dynamic range of the pixel for glucose measurement.

Voutmax = IF(T/2CM)=1.65V for single supply [1 mark ]
Capacitor size = (50um x150um) + (50um x100um)=12500um$^2$
Total capacitance = 12500um$^2$ x 4x10$^{-4}$ pF /um$^2$ =5pF [1 mark ]
$F_{max}$=IF/2$C_M V_{out}$=3.18x10$^{-6}$ / (2x5x10$^{-12}$ x 1.65) =193KHz [2 marks]

iv) Sketch a suitable schematic to interface to the pH sensor including switches to interface to more than one pixel.




[2 marks for readout of single pixel]
[2 marks for inclusion of switches and two pixels]

4. This question relates to the design of a stimulation circuit for a cortical prosthesis for vision:

a) Compare the relative merits/drawbacks between a retinal and cortical visual prosthesis. [4]

- A retinal implant is less invasive than a cortical implant (eye surgery vs brain surgery).
- The visual scene is spatiotopically mapped in the retina, this is not the case in the visual cortex.
- A cortical prosthesis can be used to treat more causes of blindness, a retinal prosthesis on the other hand requires an intact optic nerve.
- A retinal prosthesis can "host" the image detector (i.e., camera) as there is a direct optical path, whereas a cortical prosthesis would require an external camera (e.g., mounted on spectacles).

[1] each (x4)

b) Given the prosthesis will utilise penetrating electrode array (e.g. Utah), how will this influence the stimulator circuit specifications? Provide estimated values for key related specification. [6]

- Intracortical electrodes typically have a very small electrode contact area (comparable to cell/neuron size), thus high impedance, e.g., ~100kohms to 1Mohm.
- However, as the electrode is also much closer to target cells (neurons), thus charge stimulus likely to be relatively low, meaning a smaller stimulation current (e.g., <100uA), than compared to what is needed for larger electrode types.
- Voltage supply/headroom thus likely to be modest, e.g., in range 5-20V.

[2] each (x3) – [1] for stating influence and [1] for example.

c) Given the electrode material is platinum iridium (charge injection capacity ~200μC.cm$^{-2}$ per phase), estimate the maximum stimulation power consumption for a 1000-channel cortical prosthesis. State any assumptions made and describe your thought process. [8]

- Aiming for a 50% margin to the maximum capacity -> stimulus density = 100μC.cm$^{-2}$ per phase
- Assuming electrode contact surface is 500μm$^2$ -> stimulus = 100 x 500 / (10000)$^2$ = 0.5nC

Answers

- Assuming refresh rate = 25Hz and TDM/CIS stimulation strategy
- Stimulus duration (per phase) = 1/25 (refresh) x 1/1000 (pixels) x 1/2 (duty cycle) = 20μs/phase
- Assuming current-mode stimulation, stimulation current magnitude = 0.5nC/20μs = 25μA
- IR drop across tissue = I x R = 25μA x 100kohms = 2.5V
- Total power supply = 5V (assuming 2.5V across load, and 2.5V across circuit)
- Power consumption = 5V x 25μA = 125μW

[1] each (x8)

d) Describe which of the assumptions made in your answer above to (c) is likely to pose the major limitation to a visual prosthesis. [2]

- Area estimate is based on desired geometry but is not informed by threshold stimulus for activating tissue. This is just assuming half the maximum charge density into electrode of stated size. In practice however the charge injection capacity of materials such as Platinum Iridium is insufficient to achieve high resolution electrical stimulation (in both retina and cortex). This is the main unresolved challenge of visual prosthesis (and current spreading due to electrical stimulation).

5. Sequencing the human genome requires the identification of 3.2 billion bases of DNA that can be implemented using the process of pyrosequencing.

a) Explain what DNA is and describe one of its properties that allow detection of single bases for DNA sequencing.

[bookwork]
**DNA** – Deoxyribonucleic acid
To *conserve and transfer* the genetic information by replication.
**DNA double helix:** DNA organised in double stranded form
Complementary (matching) **base pairs**:
**A-T** (Adenine – Thymine) 2 hydrogen bridges
**G-C** (Guanine – Cytosine) 3 hydrogen bridges
**Nucleotides:** A nitrogenous base, a five-carbon sugar, and at least one phosphate group.
[2 points]

**Properties which allows DNA sequencing:**
The incorporation of one nucleotide leads to the production of pyrophosphate and most importantly to the release of a hydrogen ion from the hydroxyl group of the 3' terminal of the growing strand, leading to a change in the pH of the reaction due to the increase of the solution's acidity. We can detect this change in pH using **Ion-sensitive field effect transistors**.

[2 points]
[Total 4 points]

b) Explain the process of pyrosequencing, including a diagram showing the sequence of DNA bases with the signals generated for detection.

Pyrosequencing - Insert bases ATCG sequentially and see if there is a match by monitoring the change in pH.

[2 points]

Answers

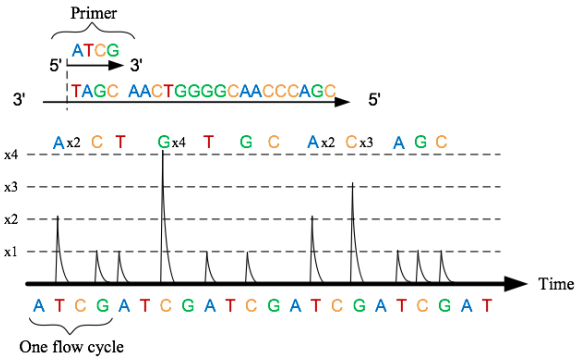



Diagram: [2 points]

[Total 4 points]

c) Name one sensor which is suitable for detecting DNA bases, drawing the structure of the device and listing the equations determining its operation.

## Ion-Sensitive Field Effect Transistors

- pH sensing MOS transistors.

Drain Current: $I_{DS(si)} = \beta(V_{GS} - V_T)^2(1 + \lambda V_{DS})$

Threshold Voltage: $V_T = V_{TH(MOS)} + V_{chem}$

Chemical potential: $V_{chem} = \gamma + 2.3\alpha U_t pH$

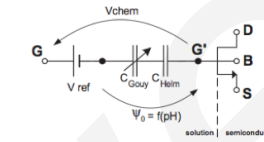
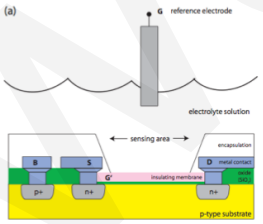
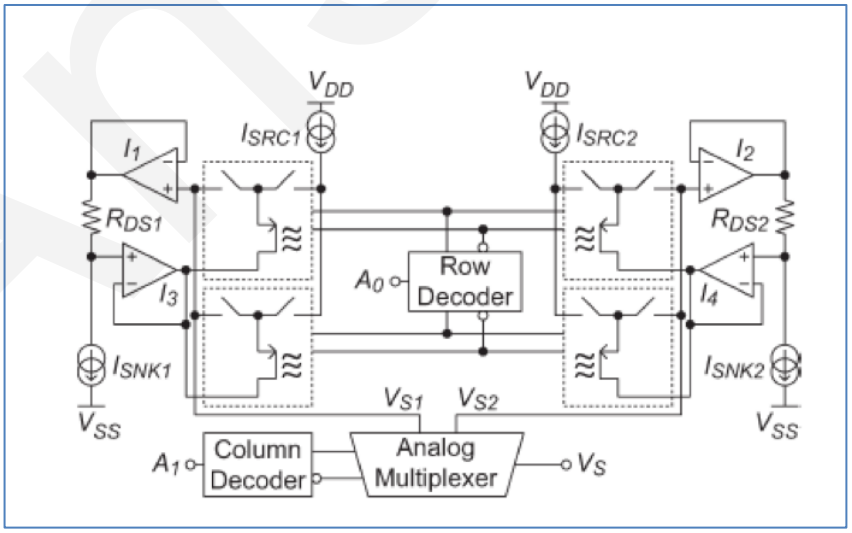



Figure 4.6: ISFET Behavioural Macromodel




[Total 4 points]

d) Draw a block level circuit schematic to show how the sensor named in part c. can be connected when scaled to millions of sensors.




[Total 5 points]

e) Explain why using this sensor has significantly reduced the cost of DNA sequencing.

ISFETs can be implemented using MOSFETs in CMOS and scaling with Moores law has allowed large scale integration to create chips with millions of sensors. We can therefore

detect lot's of DNA concurrently and this has significantly reduced the cost do DNA sequencing.

[Total 3 points]

Answers